

Credible on Paper? Evaluating the Effect of Green Banking Policy on Firms' Green Investment in Emerging Economies

Quang-Vinh Dang^{a,*}, Thi-Hong-Hanh Nguyen^b

^a*British University Vietnam, Hung Yen, Vietnam*

^b*Banking Academy Vietnam, Hanoi, Vietnam*

Abstract

More than sixty emerging and developing economies have adopted green banking guidelines through the Sustainable Banking and Finance Network (SBFN). The policy's theory of change holds that supervisory guidelines lead banks to price environmental risk, so that credit flows to firms investing in cleaner production. We evaluate two links in that chain. At firm level, using two independent World Bank Enterprise Survey samples (28,042 firms in 41 economies; 89,797 firms in 162 economies) and four estimators, bank credit is associated with 13-percentage-point higher green-practice adoption, but the association is the same in adopting and non-adopting economies (0.076 versus 0.072 in a causal forest) and does not vary with regulatory quality. At country level, a staggered-adoption event study of 217 economies finds no effect on renewable-energy share, and a CO2 association that appears to reflect which economies join. The guidelines show no detectable effect on

*Corresponding author. ORCID: Quang-Vinh Dang 0000-0002-3877-8024; Thi-Hong-Hanh Nguyen 0009-0006-7821-7404.

Email addresses: vinh.dq4@buv.edu.vn (Quang-Vinh Dang),
hanhnh1@hvn.edu.vn (Thi-Hong-Hanh Nguyen)

the margin they target. For policy making, the results suggest that green finance programmes should be judged on firm and environmental outcomes rather than on guideline adoption alone, and that guidelines aimed at banks may need complements aimed at firms, such as technical assistance or credit guarantees.

Keywords: green banking policy, policy evaluation, theory of change, access to finance, staggered difference-in-differences, causal forest

1. Introduction

Since 2012, more than sixty central banks and financial regulators in emerging and transition economies have joined the IFC-hosted Sustainable Banking and Finance Network (SBFN), issuing guidelines that direct commercial banks to price environmental risk into lending and to channel credit towards cleaner production. The network monitors this effort through policy outputs: which guidelines each member has issued, and how far it has moved along the network’s own progression framework (IFC 2025). It does not monitor whether firms change what they do. Whether a firm that can obtain a bank loan is any more likely to install pollution-control equipment or measure its own emissions because such a guideline exists is the question an evaluation of the policy has to answer.

We treat SBFN adoption as a policy intervention and evaluate it against its own theory of change (Section 2.1). That theory runs from a supervisory guideline, through banks pricing environmental risk and reallocating credit, to financed firms investing more in cleaner production, and from there to better aggregate environmental outcomes. If the chain works, two things should

18 be observable: the link between bank credit and firms' green investment
19 should be stronger where a guideline is in force, and adopting economies'
20 environmental indicators should improve after adoption. We test both.

21 Country-level studies find that green-finance policy works better where
22 governance is stronger (Chi and Yang 2023; Jia et al. 2026; Shi, Zhu, and
23 Khan 2024; Ullah, Zubir, and Ariff 2024), and firm-level studies find that
24 financing constraints shape green investment (Costa et al. 2024; De Haas et
25 al. 2025; Ghosh and Dutta 2022). Neither literature has tested, on firm-level
26 data and across a broad cross-section of adopting economies, whether green
27 banking guidelines strengthen the credit-to-green channel they are designed
28 to activate.

29 We use two independently assembled World Bank Enterprise Survey (WBES)
30 samples: a 41-economy primary sample of 28,042 firms surveyed in 2018–
31 2020 with the Bank's full Green Economy module, and a 162-economy global
32 sample of 89,797 firms surveyed in 2021–2026 that extends to high-income
33 economies outside the SBFN's remit. Because SBFN status is constant within
34 a country in a single survey wave, we combine a pooled logit benchmark with
35 a Bayesian hierarchical model, a causal forest, and, on the larger sample,
36 country fixed effects. Across all four estimators the answer is the same, and it
37 is precise: in the causal forest the credit effect is 0.076 in adopting economies
38 and 0.072 in non-adopting ones. A Callaway–Sant'Anna staggered-adoption
39 event study of 217 economies then tests the aggregate outcome directly.

40 The paper contributes to the evaluation of green finance policy in four
41 ways. It evaluates SBFN-style adoption against firm-level outcomes rather
42 than against the policy-stage indicators the network reports or the bank-level

lending data that dominate the green-credit literature. It tests the theory of change at two points, with different data, units of analysis and identification strategies. It replicates the firm-level result on an independent sample with a different outcome and country fixed effects. And because the estimates are precise rather than merely insignificant, it can report a genuine absence of effect on the targeted margin, the kind of null result that matters for programme design but is rarely published.

Section 2 describes the SBFN framework and its theory of change. Section 3 develops the hypotheses, Section 4 sets out data and methods, Section 5 evaluates the firm-level credit channel and Section 6 the aggregate outcome. Section 7 draws lessons for policy and evaluation, and Section 8 concludes.

2. Green banking policy and its theory of change

The SBFN, founded in 2012 as the Sustainable Banking Network and hosted by the International Finance Corporation, is the main forum through which financial regulators and banking associations in emerging and developing economies coordinate national sustainable-finance policy. Its roughly seventy member institutions come almost entirely from Africa, Asia, Latin America, the Middle East and non-EU Europe (IFC 2025); none of the sixteen EU member states in our primary sample is a member.

Membership is the network’s coarsest measure of adoption. Its Multi-Tiered Progress Framework places each member on a progression from ‘Commitment’ or ‘Formulating’, through ‘Developing’, to ‘Advancing’ and ‘Consolidating’. Our sample spans that whole range: Mongolia and Morocco had reached ‘Advancing’ by their 2019 surveys, whereas Jordan and the Kyrgyz

67 Republic were still at ‘Commitment’. Several economies, including Kaza-
68 khstan, Serbia, Ukraine and most of the Western Balkans, joined only after
69 their survey was fielded. We code these as non-adopters, because what mat-
70 ters is the policy environment in force when a firm answered the survey.

71 *2.1. The policy’s theory of change and what this evaluation tests*

72 The case that SBFN adoption should change firm behaviour rests on a
73 supervisory mechanism with five links:

- 74 1. *Adoption.* A banking regulator or banking association issues environmental-
75 and-social risk management guidelines.
- 76 2. *Bank response.* Supervised banks face a compliance incentive to price
77 environmental risk into lending and, in more advanced frameworks, to
78 report a green-lending share.
- 79 3. *Credit allocation.* Credit flows more readily to firms undertaking cleaner
80 investment and less readily to heavy polluters.
- 81 4. *Firm response.* Firms with bank credit invest more in green practices
82 than they would without the guideline, so the credit-to-green link is
83 stronger where the guideline is in force (H2), and stronger still where
84 the supervisor can enforce it (H3).
- 85 5. *Aggregate outcome.* Adopting economies’ environmental indicators im-
86 prove relative to non-adopters after adoption.

87 Evidence on China’s green credit guidelines documents links 2 to 4 at
88 bank and firm level (Dai et al. 2025; Gao et al. 2022; Huang, Gao, and Jia
89 2023; Li et al. 2023), and studies of Bangladesh and Vietnam point the same
90 way (Khairunnessa, Vazquez-Brust, and Yakovleva 2021; Nguyen and Phan

2025). Whether the chain holds across a heterogeneous set of SBFN adopters is untested. This paper tests link 4 at firm level (Section 5) and link 5 at country level (Section 6). It does not observe links 2 and 3: the Enterprise Surveys record whether a firm holds bank credit, not the terms banks offered or the green share of their lending. A null at link 4 therefore shows that the chain does not deliver its intended firm-level effect, but cannot on its own say where it breaks.

3. Hypotheses

Green investments such as energy-management systems, on-site renewable generation and waste-minimisation retrofits carry a large up-front cost against savings realised over several years, the profile of investment most sensitive to external finance. Firm-level evidence is consistent with this. Credit-constrained Indian manufacturers are less likely to adopt cleaner production (Ghosh and Dutta 2022); financial and managerial barriers jointly constrain green investment across countries (De Haas et al. 2025; Kalantzis, Schweiger, and Dominguez 2022); financing constraints blunt firms' response to environmental policy in OECD and Vietnamese data (Costa et al. 2024; Truong 2026); and firms with better credit access sustain green management through shocks (Aristei and Gallo 2023). The WBES Green Economy module has been used to study such behaviour across many countries (Asif et al. 2025; Phan 2026). The first-order proposition is:

H1. Firms with access to formal bank credit (a credit line or loan) are more likely to have adopted green production practices than otherwise similar firms without it.

115 H1 describes a channel, not a policy. The premise of SBFN-style guide-
116 lines is that a supervisory mandate should widen the gap in green investment
117 between firms with and without bank credit, because banks in adopting ju-
118 risdictions have an incentive to seek out and price favourably exactly the
119 borrowers making such investments. Most supporting evidence comes from
120 single-country studies of China’s unusually well-resourced regime (Dai et al.
121 2025; Gao et al. 2022). Whether the same moderation holds across the
122 far more heterogeneous, and often less advanced, set of SBFN adopters is
123 untested:

124 **H2.** The association between bank credit and green-practice
125 adoption (H1) is stronger in economies that had adopted an
126 SBFN framework by the survey year than in economies that had
127 not.

128 A guideline is only as effective as the supervisor behind it. The law-
129 and-finance and state-capacity literatures show that formally identical rules
130 produce different outcomes depending on the institutions that enforce them
131 (Acemoglu, Johnson, and Robinson 2001; La Porta et al. 1998; Levine 2005).
132 Where examination authority, sanctions and disclosure enforcement are weak,
133 an SBFN framework may exist on paper without changing lending. Country-
134 level evidence fits this view: green finance improves environmental outcomes
135 only where corruption is controlled (Shi, Zhu, and Khan 2024), China’s guide-
136 lines work best where environmental governance is strong (Chi and Yang
137 2023), and governance strengthens the green-credit-to-sustainable-production
138 link in BRICS economies (Jia et al. 2026). Each result implies an untested

139 firm-level prediction. Proxying regulatory capacity by the Worldwide Gov-
140 ernance Indicators (WGI) Regulatory Quality estimate, we hypothesise a
141 three-way interaction:

142 **H3.** The moderation in H2 increases with regulatory quality:
143 SBFN adoption strengthens the credit-to-green channel most where
144 the supervisor has the capacity to make the guideline credible.

145 *3.1. An auxiliary, exploratory hypothesis*

146 Financial constraints bind environmental investment most tightly in smaller
147 firms (Ullah 2025), consistent with a fixed cost that is proportionately heavier
148 for them even once financed. We did not pre-specify a firm-size hypothesis. It
149 emerged from the causal forest (Section 5.4), and we report it as exploratory:

150 **H4 (exploratory).** Heterogeneity in the credit-to-green chan-
151 nel is more attributable to firm size than to either country-level
152 moderator.

153 **4. Data and methods**

154 *4.1. Data*

155 The WBES are stratified random-sample surveys of formal, non-agricultural
156 private firms with at least five employees. Our primary sample is the Bank’s
157 harmonised file for 41 Europe and Central Asia and Middle East and North
158 Africa economies surveyed with an identical questionnaire in 2018–2020, the
159 wave in which the Green Economy module was fielded in full: 28,042 firms
160 (Supplementary Table S2). Six further economies (Bangladesh 2022, India

161 2022, Indonesia 2023, Peru 2023, the Philippines 2023 and Timor-Leste 2021;
162 15,556 firms) fielded a shortened, partly randomised version of the module.
163 Because item wording differs, we do not pool them, and use them only for a
164 narrower check (Section 5.5).

165 The global sample comes from the Bank’s standardised cross-country
166 database, which harmonises core variables across 471 country-years and 292,314
167 firms. In it, 160 country-years fielded in 2021–2026, including high-income
168 economies such as Australia, Canada, Germany and Japan, carry CO2-
169 emissions monitoring alongside the standard access-to-finance questions. Adding
170 Bangladesh 2022 and Indonesia 2023 from our own extraction of the country
171 files gives 162 economies and 100,115 firms, of which 89,797 have complete
172 data. Controls come from the Bank’s Productivity Database, merged by firm
173 identifier.

174 SBFN status, join year and progression stage for the primary sample
175 are compiled from the network’s Global Progress Reports (2018, 2019, 2021,
176 2023–2024), country addenda and data portal, cross-checked against IFC
177 accession announcements. For the global sample we use the portal’s mem-
178 ber roster (67 members). In both, an economy counts as an adopter only
179 if it joined before its survey year. Six CEMAC economies and four East-
180 ern Caribbean economies appear on the roster under regional entries, which
181 we read as covering their common banking supervisors; these are inferences
182 rather than per-country listings. Regulatory quality is the WGI estimate
183 for the survey year (or the latest earlier year); WGI does not cover Taiwan,
184 which is dropped.

185 A staggered difference-in-differences design around each economy’s join

186 date is not feasible at firm level. The Green Economy module has been fielded
 187 to most economies only once, and economies surveyed twice by the core
 188 instrument did not repeat a comparable module, so no consistent outcome is
 189 observed before and after adoption. We therefore run the staggered design
 190 at country level (Section 6) and treat the firm-level cross-section as a test of
 191 link 4.

192 4.2. Variables

193 The outcome, *green practice adoption*, equals one if a firm adopted at
 194 least one of seven practices in the previous three years: climate-friendly on-
 195 site energy generation, an energy-management system, waste minimisation or
 196 recycling, air-pollution control, other pollution control, any energy-efficiency
 197 measure, or on-site renewable energy. The mean of the seven items serves as
 198 a continuous index for robustness. *Bank credit* equals one if the firm holds a
 199 credit line or loan from a formal financial institution. *SBFN member* equals
 200 one if the economy had adopted an SBFN framework by the survey year, and
 201 *regulatory quality* is the WGI estimate, standardised within the sample for
 202 the hierarchical and machine-learning models. Controls are firm size (small,
 203 medium or large), broad sector, log sales, exporter status and any foreign
 204 ownership. Supplementary Table S1 gives the WBES code and source of
 205 every variable.

206 4.3. Estimation

207 **Stage 1: pooled logit benchmark.** We estimate

$$\Pr(\text{Green}_{i,c} = 1) = \Lambda\left(\beta_1 \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + \delta_{s(i)}\right) \quad (1)$$

for firm i in country c , with firm controls $\mathbf{X}_{i,c}$, sector fixed effects $\delta_{s(i)}$ and country-clustered standard errors. A country fixed effect would absorb SBFN $_c$, so β_2 would be unidentified. The interaction β_3 would remain identified, because credit varies within countries, but only from the contrast between eight adopter economies and thirty-three non-adopters, and the level difference between them would be lost. We therefore estimate β_2 and β_3 without country fixed effects, and add a triple interaction with regulatory quality to test H3 directly (column M3).

Stage 2: Bayesian hierarchical model. Country enters as a random effect, with a random slope on credit:

$$\text{Green}_{i,c} \sim \text{Bernoulli}(p_{i,c}) \quad (2)$$

$$\begin{aligned} \text{logit}(p_{i,c}) = & (\beta_1 + u_{1,c}) \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) \\ & + \beta_4 (\text{Credit}_{i,c} \times \text{RegQuality}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + (\alpha + u_{0,c}) \end{aligned} \quad (3)$$

$$\begin{pmatrix} u_{0,c} \\ u_{1,c} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (4)$$

estimated by the No-U-Turn Sampler in PyMC (Salvatier, Wiecki, and Fonnesbeck 2016). Partial pooling keeps the country-level main effects identified while giving more defensible inference than cluster-robust standard errors over only 41 clusters (Cameron and Miller 2015; Gelman and Hill 2007). β_3 tests H2; β_4 tests whether regulatory quality alone moderates the credit-green link. Neither is a test of H3, which needs a three-way term that eight adopter economies cannot support under partial pooling. H3 is tested by Stage 1's column M3 and by the saturated global-sample model.

Stage 3: causal forest. To avoid imposing a linear interaction, we

227 estimate a causal forest with double machine learning (Athey, Tibshirani,
 228 and Wager 2019; Chernozhukov et al. 2018; Wager and Athey 2018), using
 229 gradient-boosted nuisance models and firm size, export status, SBFN status
 230 and regulatory quality as effect modifiers. Its identifying assumption is the
 231 same unconfoundedness on observables as Stages 1 and 2, so its conditional
 232 average treatment effects (CATEs) are flexible conditional associations, not
 233 effects identified from exogenous variation.

234 **Stage 4: global sample.** We re-estimate the model on the global sample
 235 with CO2-emissions monitoring as the outcome. With 159 country clusters,
 236 country fixed effects become usable: they absorb the SBFN and regulatory-
 237 quality main effects but leave their interactions with credit identified. We
 238 report this alongside a saturated specification without fixed effects that re-
 239 tains the three-way term.

240 5. Evaluating the firm-level credit channel

241 5.1. Descriptive patterns

242 Firm counts in the primary sample range from 150 (Montenegro) to 3,075
 243 (Egypt), and the unconditional adoption rate from 30.3% (Egypt) to 90.1%
 244 (Malta). Eight of the 41 economies had adopted an SBFN framework by
 245 their survey year: Egypt, Georgia, Jordan, the Kyrgyz Republic, Mongolia,
 246 Morocco, Tunisia and Türkiye (Figure 1). Bank credit correlates with the
 247 adoption index ($r = 0.18$) but not with firms' perceived finance obstacle
 248 ($r = -0.02$). Figure 2 previews the main result. Plotting adoption against
 249 regulatory quality separately for members and non-members gives two lines
 250 of similar slope about 15–20 percentage points apart: a level difference, with

251 no difference in slope.

252 5.2. Pooled logit benchmark

253 Table 1 supports H1. Firms with a bank credit line are more likely to have
254 adopted a green practice ($b = 0.571$, $p < 0.001$), with an average marginal
255 effect of 12.95 percentage points (95% CI [7.7, 18.2]), about the same as
256 moving from a small to a large firm. Column M2 shows a large negative
257 SBFN main effect ($b = -0.991$, $p < 0.001$) and a small, insignificant credit $\times$
258 SBFN interaction ($b = 0.049$, $p = 0.773$), contrary to H2. Column M3 tests
259 H3: the credit $\times$ SBFN $\times$ regulatory-quality term is small, of the wrong
260 sign and insignificant ($b = -0.104$, $se = 0.272$, $p = 0.702$). Regulatory
261 quality's own association with adoption is positive but imprecise ($b = 0.278$,
262 $p = 0.110$).

263 5.3. Bayesian hierarchical model

264 The hierarchical model (Table 2; four chains, all $\hat{R} \leq 1.01$, bulk effective
265 sample sizes 797–2,270; Supplementary Section S7) gives H2 a fair hear-
266 ing without relying on 41-cluster standard errors. The credit effect survives
267 (posterior mean 0.33, 89% interval [0.18, 0.49]). SBFN membership lowers
268 baseline adoption (-0.57 , $[-1.00, -0.09]$) and regulatory quality raises it
269 (0.34 , $[0.16, 0.53]$), reproducing the level pattern in Figure 2. Neither inter-
270 action differs from zero: credit $\times$ SBFN is 0.072 ($[-0.26, 0.40]$) and credit $\times$
271 regulatory quality is 0.004 ($[-0.13, 0.14]$), a tight interval around zero rather
272 than merely a wide one.

273 5.4. *Causal forest*

274 The causal forest (Table 3, Figure 3) imposes no functional form on mod-
275 eration. Its average effect of credit is 0.073 (95% CI $[-0.040, 0.186]$), smaller
276 and less precise than the logit marginal effect, as expected of a cross-fitted,
277 doubly robust estimator. The CATEs are flat across every institutional cut:
278 0.072 in non-adopting and 0.076 in adopting economies, and 0.075, 0.070 and
279 0.075 across the low, middle and high regulatory-quality terciles.

280 Firm size accounts for 75.6% of the forest’s explained heterogeneity, against
281 19.9% for regulatory quality, 2.6% for SBFN status and 1.9% for exporter
282 status, consistent with H4. But the size strata do not show a monotonic gra-
283 dient (0.080 for small, 0.075 for medium and 0.057 for large firms, all with
284 overlapping intervals that include zero; Supplementary Table S3). We can
285 say that heterogeneity sits at firm level rather than with country institutions,
286 but not in which direction it runs.

287 5.5. *Robustness*

288 The result holds with the continuous adoption index estimated by OLS,
289 and separately for manufacturing and services (Supplementary Table S5). In
290 four of the six supplementary economies, an overdraft facility is associated
291 with higher waste-minimisation adoption ($b = 0.422$, $p < 0.001$, $N = 9,038$;
292 Supplementary Table S4), so the credit channel is not specific to the primary
293 region. One interaction is nominally significant: with overdraft access in
294 place of a credit line, the overdraft $\times$ SBFN term is negative ($b = -0.346$, $p =$
295 0.031), the opposite of H2’s prediction. It is one of thirteen interaction tests
296 under frequentist estimators, and after a Benjamini–Hochberg correction its

297 p -value rises to 0.403; no test in the family survives at a false discovery rate
298 of 0.10 (Supplementary Table S6).

299 *5.6. Replication on the global sample*

300 The global sample differs from the primary one in outcome, period and
301 economies, with little overlap (Figure 4). Without country fixed effects (Ta-
302 ble 4, M1), credit is associated with emissions monitoring ($b = 0.198$, $p =$
303 0.039 ; 89,797 firms, 159 countries), while the SBFN main effect ($b = -0.171$,
304 $p = 0.353$), the credit $\times$ SBFN interaction ($b = 0.278$, $p = 0.430$) and
305 the regulatory-quality interactions are insignificant, including the three-way
306 term this saturated model retains. With country fixed effects (M2), credit
307 is more precisely estimated ($b = 0.249$, $p < 0.001$), and neither the credit $\times$
308 SBFN ($b = 0.506$, $p = 0.205$) nor the credit $\times$ regulatory-quality interaction
309 ($b = -0.044$, $p = 0.346$) is significant. Figure 5 collects every estimate. The
310 credit effect is positive in both samples and under every estimator, and every
311 interaction interval includes zero.

312 **6. Evaluating the aggregate outcome: a staggered-adoption event** 313 **study**

314 The firm-level evidence addresses link 4. Link 5, the aggregate outcome
315 the policy is ultimately meant to move, can be tested directly using SBFN’s
316 staggered join dates, because country-level outcomes are observed every year.

317 *6.1. Data and design*

318 We build a country-year panel of 217 economies for 2000–2024. Treatment
319 timing is each economy’s SBFN join year from the portal roster, which agrees

with our global-sample coding for all 61 overlapping economies. This gives 68 ever-treated economies (join years 2012–2024) and 149 never-treated ones. Outcomes from the World Development Indicators are renewable energy as a share of final energy consumption and CO2 emissions per capita; log GDP per capita (constant 2015 US dollars) is a covariate, because adopters are disproportionately fast-growing, industrialising economies.

We use the Callaway and Sant’Anna (2021) estimator, which estimates group-time average treatment effects by adoption cohort and aggregates them into an overall average treatment effect on the treated (ATT) and an event-study profile, with never-treated economies as controls and 999 bootstrap replications. Two-way fixed effects are biased under staggered adoption (Goodman-Bacon 2021), so we report them only to show the size of that bias. For each outcome we estimate an unconditional and a doubly robust specification with log GDP per capita, and check the latter against a not-yet-treated control group.

6.2. Results

Renewable-energy share shows no effect (Table 5): -1.192 (95% CI $[-2.763, 0.380]$) unconditionally and 0.674 ($[-1.122, 2.470]$) with the GDP control, and the event-study profile is flat on both sides of adoption (Figure 6). CO2 emissions per capita show a positive association, but three features argue against a causal reading. It falls by about 45%, from 0.631 to 0.349 , once GDP per capita is controlled for. Its event-study profile rises steadily from near zero at adoption to a peak nine to twelve years later, which looks like adopters already being on a steeper emissions path rather than like a policy effect. And two-way fixed effects give 0.901 , more than double the doubly

robust estimate. Both doubly robust results are essentially unchanged with not-yet-treated controls (0.313 for CO2 and 0.653 for renewables). The design finds no evidence that adoption improves an economy’s environmental trajectory, and its one significant result is better explained by which economies choose to join.

7. Discussion

In plain terms, a firm with a bank credit line is about 13 percentage points more likely to have adopted a green practice than a similar firm without one. This is an association, not a causal effect. The gap is the same whether or not the firm’s country has adopted an SBFN framework, and at every level of regulatory quality. Firms in adopting economies are instead 15–20 percentage points less likely to have adopted any green practice at all: a difference in level, not in slope. Because the two firm-level samples share few economies, periods or outcome items, and the event study uses different data, units and identification, this convergence is unlikely to be an artefact of one sample or estimator.

The result complicates a green-credit literature built largely on China’s mature guidelines (Dai et al. 2025; Huang, Gao, and Jia 2023; Li et al. 2023). Our adopters span every stage of SBFN’s progression framework, and across that range we find no sign that formal adoption changes how credit translates into green investment. ‘No interaction detected’ is not the same as ‘no complementarity exists’. A linear interaction may miss a threshold effect, given that SBFN’s framework is graded and our coding is binary, and the WGI may not capture banking-supervisory capacity. We consider our reading

369 the best supported by the evidence, but it does not rule these alternatives
370 out.

371 The one nominally significant interaction, a negative overdraft $\times$ SBFN
372 term, is at least consistent with guidelines steering credit towards term
373 lending rather than towards working capital, although it does not survive
374 multiple-testing correction. More generally, if smaller firms cannot absorb
375 the fixed costs of green investment even when financed, a guideline aimed at
376 bank lending acts on the wrong margin however credible it is.

377 *7.1. Lessons for policy and for evaluation*

378 First, monitoring adoption is not monitoring effect. SBFN's progress
379 framework scores members on the guidelines they issue and the stage they
380 reach, and on our evidence being further along it does not coincide with any
381 change in the firm-level margin the guidelines target. Programmes of this
382 kind need outcome indicators alongside output indicators, at a minimum
383 the share of bank credit financing green investment and firm-level adoption
384 measured before and after a guideline takes effect.

385 Second, the data needed for a proper firm-level evaluation do not yet exist.
386 Re-fielding a comparable Green Economy module in the same economies
387 would allow the within-country before-and-after comparison this paper could
388 run only at country level, and would let future evaluations locate where in
389 the theory of change the chain breaks.

390 Third, bank-side guidelines may need firm-side complements, such as
391 technical assistance, matching grants or credit guarantees for smaller firms, if
392 the binding constraint lies with firms rather than with banks. The firm-size
393 evidence here points that way but is exploratory.

394 8. Conclusion

395 We evaluated SBFN green banking guidelines against their own theory of
396 change. Bank credit is robustly associated with green-practice adoption, but
397 that association is no stronger in adopting economies and does not depend on
398 regulatory quality, on either of two independent firm-level samples and under
399 four estimators. A staggered-adoption event study finds no improvement in
400 adopters' aggregate environmental outcomes. The policy implication is not
401 that green banking guidelines are ineffective in every respect, but that our
402 evidence gives no support to treating them as a lever that widens the green-
403 investment gap between financed and unfinanced firms.

404 The evaluation has limits. The firm-level samples are cross-sections and
405 cannot rule out reverse causality or unobserved country characteristics. SBFN
406 membership is a binary proxy for a graded process, and WGI regulatory qual-
407 ity is a broad governance measure rather than a supervisory one. The global
408 sample's outcome is a single item. The design tests the intensive margin of
409 the credit channel, not whether guidelines change who obtains credit, which
410 our measure of realised access cannot observe, since some firms decline formal
411 credit altogether (Jin 2026). And the event study, an aggregate test netted
412 against one covariate, cannot exclude a small confounded relationship.

413 Acknowledgements

414 We thank the Enterprise Analysis Unit of the Development Economics
415 Global Indicators Department of the World Bank Group for making the
416 Enterprise Surveys data available.

417 **Disclosure statement**

418 The authors report there are no competing interests to declare.

419 **Funding**

420 No funding was obtained for this research.

421 **Declaration of generative AI use**

422 The authors used generative AI tools to assist with drafting, editing and
423 restructuring the manuscript text and with writing and checking analysis
424 code. The authors reviewed and edited all AI-assisted output and take full
425 responsibility for the content of the article.

426 **Notes on contributors**

427 **Quang-Vinh Dang** [BIOGRAPHICAL NOTE TO BE SUPPLIED BY
428 THE AUTHOR, max. 200 words.]

429 **Thi-Hong-Hanh Nguyen** [BIOGRAPHICAL NOTE TO BE SUPPLIED
430 BY THE AUTHOR, max. 200 words.]

431 **Data availability statement**

432 The firm-level microdata are available from the World Bank Enterprise
433 Surveys (enterprisesurveys.org; free registration required) under the Surveys'
434 terms of use. The country-level datasets constructed for this paper and the
435 full analysis code are available in a public repository at <https://github.com/vinhqdang/green-credit-substitution>. Supplementary material ac-
436 companies the article.

438 **References**

- 439 Acemoglu, Daron, Simon Johnson, and James A. Robinson. 2001. “The Colo-
440 nial Origins of Comparative Development: An Empirical Investigation.”
441 *American Economic Review* 91 (5): 1369–1401. [https://doi.org/10.](https://doi.org/10.1257/aer.91.5.1369)
442 [1257/aer.91.5.1369](https://doi.org/10.1257/aer.91.5.1369)
- 443 Aristei, David, and Manuela Gallo. 2023. “Green Management, Access to
444 Credit, and Firms’ Vulnerability to the COVID-19 Crisis.” *Small Business*
445 *Economics*: 1–33. <https://doi.org/10.1007/s11187-023-00759-1>
- 446 Asif, Mohd, Amit Dhamija, Wasim Khan, Amit Johri, and Mohammad
447 Wasiq. 2025. “Factors Affecting Adoption of Green Practices by Food
448 Firms: Evidences from the World Bank Enterprises Survey.” *Clean Tech-*
449 *nologies and Environmental Policy* 27 (9): 4841–4858. [https://doi.org/](https://doi.org/10.1007/s10098-024-03122-4)
450 [10.1007/s10098-024-03122-4](https://doi.org/10.1007/s10098-024-03122-4)
- 451 Athey, Susan, Julie Tibshirani, and Stefan Wager. 2019. “Generalized Ran-
452 dom Forests.” *The Annals of Statistics* 47 (2): 1148–1178. [https://doi.](https://doi.org/10.1214/18-AOS1709)
453 [org/10.1214/18-AOS1709](https://doi.org/10.1214/18-AOS1709)
- 454 Callaway, Brantly, and Pedro H. C. Sant’Anna. 2021. “Difference-In-
455 Differences with Multiple Time Periods.” *Journal of Econometrics* 225 (2):
456 200–230. <https://doi.org/10.1016/j.jeconom.2020.12.001>
- 457 Cameron, A. Colin, and Douglas L. Miller. 2015. “A Practitioner’s Guide to
458 Cluster-Robust Inference.” *Journal of Human Resources* 50 (2): 317–372.
459 <https://doi.org/10.3368/jhr.50.2.317>

- 460 Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo,
461 Christian Hansen, Whitney Newey, and James Robins. 2018. “Double/
462 ble/debiased Machine Learning for Treatment and Structural Parameters.”
463 *The Econometrics Journal* 21 (1): C1–C68. [https://doi.org/10.1111/](https://doi.org/10.1111/ectj.12097)
464 [ectj.12097](https://doi.org/10.1111/ectj.12097)
- 465 Chi, Yi, and Yang Yang. 2023. “Green Finance and Green Transition by En-
466 terprises: An Exploration of Market-Oriented Governance Mechanisms.”
467 *Borsa Istanbul Review* 23 (3): 628–646. [https://doi.org/10.1016/j.](https://doi.org/10.1016/j.bir.2023.01.003)
468 [bir.2023.01.003](https://doi.org/10.1016/j.bir.2023.01.003)
- 469 Costa, Helia, Lilas Demmou, Guido Franco, and Stefan Lamp. 2024. “The
470 Role of Financing Constraints and Environmental Policy on Green In-
471 vestment.” *Economics Letters* 239: 111741. [https://doi.org/10.1016/](https://doi.org/10.1016/j.econlet.2024.111741)
472 [j.econlet.2024.111741](https://doi.org/10.1016/j.econlet.2024.111741)
- 473 Dai, Qi, Jiaxuan He, Zhen Guo, Yuting Zheng, and Yue Zhang. 2025. “Green
474 Finance for Sustainable Development: Analyzing the Effects of Green
475 Credit on High-Polluting Firms’ Environmental Performance.” *Humanities and Social Sciences Communications* 12 (1): 854. [https://doi.org/](https://doi.org/10.1057/s41599-025-05218-8)
476 [10.1057/s41599-025-05218-8](https://doi.org/10.1057/s41599-025-05218-8)
- 478 De Haas, Ralph, Ralf Martin, Mirabelle Muûls, and Helena Schweiger. 2025.
479 “Managerial and Financial Barriers to the Green Transition.” *Management*
480 *Science* 71 (4): 2890–2921. <https://doi.org/10.1287/mnsc.2023.00772>
- 481 Gao, Dawei, Xingjian Mo, Kun Duan, and Yingjie Li. 2022. “Can Green

- 482 Credit Policy Promote Firms' Green Innovation? Evidence from China.”
483 *Sustainability* 14 (7): 3911. <https://doi.org/10.3390/su14073911>
- 484 Gelman, Andrew, and Jennifer Hill. 2007. *Data Analysis Using Regres-*
485 *sion and Multilevel/hierarchical Models*. Cambridge: Cambridge Univer-
486 sity Press. <https://doi.org/10.1017/CB09780511790942>
- 487 Ghosh, Debasree, and Mousumi Dutta. 2022. “Environmental Behaviour
488 Under Credit Constraints – Evidence from Panel of Indian Manufactur-
489 ing Firms.” *Structural Change and Economic Dynamics* 63: 490–500.
490 <https://doi.org/10.1016/j.strueco.2022.07.004>
- 491 Goodman-Bacon, Andrew. 2021. “Difference-In-Differences with Variation in
492 Treatment Timing.” *Journal of Econometrics* 225 (2): 254–277. <https://doi.org/10.1016/j.jeconom.2021.03.014>
- 494 Huang, Zhen, Ning Gao, and Ming Jia. 2023. “Green Credit and Its Obstacles:
495 Evidence from China’s Green Credit Guidelines.” *Journal of Corporate Fi-*
496 *nance* 82: 102441. <https://doi.org/10.1016/j.jcorpfin.2023.102441>
- 497 IFC (International Finance Corporation). 2025. *Sustain-*
498 *able Banking and Finance Network Global Progress Re-*
499 *port 2025*. Washington, DC: World Bank Group. [https://documents1.worldbank.org/curated/en/099634202132617625/](https://documents1.worldbank.org/curated/en/099634202132617625/pdf/IDU-3c06eadd-dd50-446d-a6d0-3fe3907bf5d5.pdf)
500 [pdf/IDU-3c06eadd-dd50-446d-a6d0-3fe3907bf5d5.pdf](https://documents1.worldbank.org/curated/en/099634202132617625/pdf/IDU-3c06eadd-dd50-446d-a6d0-3fe3907bf5d5.pdf)
- 502 Jia, Saitao, Umar Farooq, Abdulrahman Alomair, and Abdullah A.
503 Aljughaiman. 2026. “Credit with a Climate Purpose: Does Green Credit

- 504 Policy Accelerate Sustainable Production in BRICS?" *Borsa Istanbul Re-*
505 *view* 26 (3): 100817. <https://doi.org/10.1016/j.bir.2026.100817>
- 506 Jin, Wenwen. 2026. "Choosing Not to Borrow: Imprinting Effects of Informal-
507 mality on Firms' Credit Self-Rationing." *Borsa Istanbul Review* 26 (2):
508 100785. <https://doi.org/10.1016/j.bir.2026.100785>
- 509 Kalantzis, Fotios, Helena Schweiger, and Sara Dominguez. 2022. *Green In-*
510 *vestment by Firms: Finance or Climate Driven?* EBRD Working Pa-
511 per 268. London: European Bank for Reconstruction and Development.
512 <https://doi.org/10.2139/ssrn.4132545>
- 513 Khairunnessa, Fatima, Diego A. Vazquez-Brust, and Natalia Yakovleva.
514 2021. "A Review of the Recent Developments of Green Banking in
515 Bangladesh." *Sustainability* 13 (4): 1904. <https://doi.org/10.3390/su13041904>
- 517 La Porta, Rafael, Florencio Lopez-de-Silanes, Andrei Shleifer, and Robert
518 Vishny. 1998. "Law and Finance." *Journal of Political Economy* 106 (6):
519 1113–1155. <https://doi.org/10.1086/250042>
- 520 Levine, Ross. 2005. "Finance and Growth: Theory and Evidence." In
521 *Handbook of Economic Growth*, vol. 1A, edited by Philippe Aghion and
522 Steven Durlauf, 865–934. Amsterdam: Elsevier. [https://doi.org/10.](https://doi.org/10.1016/S1574-0684(05)01012-9)
523 [1016/S1574-0684\(05\)01012-9](https://doi.org/10.1016/S1574-0684(05)01012-9)
- 524 Li, Chao, Xiao Feng, Xin Li, and Yang Zhou. 2023. "Effect of Green
525 Credit Policy on Energy Firms' Growth: Evidence from China." *Eco-*

- 526 *nomic Research-Ekonomika Istraživanja* 36 (2). [https://doi.org/10.](https://doi.org/10.1080/1331677X.2023.2177701)
527 1080/1331677X.2023.2177701
- 528 Nguyen, Minh Phuong, and Anh Phan. 2025. “Impact of Green Credit Policy
529 on the Operation of Vietnamese Commercial Banks: An Empirical Study
530 Using Difference-In-Differences Model.” *PLOS One* 20 (10): e0333759.
531 <https://doi.org/10.1371/journal.pone.0333759>
- 532 Phan, Quang Thanh. 2026. “The Impact of Green Economy Practices
533 on Firm Innovation Using Data from the World Bank Enterprise Sur-
534 veys.” *Discover Sustainability* 7 (1): 848. [https://doi.org/10.1007/](https://doi.org/10.1007/s43621-026-03226-5)
535 s43621-026-03226-5
- 536 Salvatier, John, Thomas V. Wiecki, and Christopher Fonnesbeck. 2016.
537 “Probabilistic Programming in Python Using PyMC3.” *PeerJ Computer*
538 *Science* 2: e55. <https://doi.org/10.7717/peerj-cs.55>
- 539 Shi, Yacong, Qiju Zhu, and Muhammad Atif Khan. 2024. “The Efficacy of
540 Green Finance for Environmental Sustainability: Does Control of Cor-
541 ruption Makes a Difference?” *Borsa Istanbul Review* 24 (6): 1179–1189.
542 <https://doi.org/10.1016/j.bir.2024.07.002>
- 543 Truong, Duc Dinh. 2026. “Does Access to Finance Condition Firms’ Green In-
544 vestment Responses to Environmental Pressure? Evidence from Vietnam.”
545 *PLOS One* 21 (5): e0341960. [https://doi.org/10.1371/journal.pone.](https://doi.org/10.1371/journal.pone.0341960)
546 0341960
- 547 Ullah, Barkat. 2025. “Corporate Environmental Responsibility and Finan-

- 548 cial Constraints for Unlisted SMEs.” *Global Finance Journal* 67: 101176.
549 <https://doi.org/10.1016/j.gfj.2025.101176>
- 550 Ullah, Wasim, Ahmad Shauqi Mohamad Zubir, and Akmalia Mohamad Ariff.
551 2024. “Exploring the Moderating Effect of Regulatory Quality on the Rela-
552 tionship Between Financial Development and Economic Growth/economic
553 Volatility for Developed and Developing Countries.” *Borsa Istanbul Review*
554 24 (5): 934–944. <https://doi.org/10.1016/j.bir.2024.04.015>
- 555 Wager, Stefan, and Susan Athey. 2018. “Estimation and Inference of Hetero-
556 geneous Treatment Effects Using Random Forests.” *Journal of the Amer-
557 ican Statistical Association* 113 (523): 1228–1242. [https://doi.org/10.](https://doi.org/10.1080/01621459.2017.1319839)
558 [1080/01621459.2017.1319839](https://doi.org/10.1080/01621459.2017.1319839)

559 List of Figures

560	1	Green practice adoption and SBFN policy membership, by	
561		economy.	28
562	2	Green adoption vs. regulatory quality, by SBFN status.	29
563	3	Causal forest conditional average treatment effects, by subgroup.	30
564	4	Combined sample coverage: 166 economies (41 primary + 162	
565		global, net of overlap), shaded by SBFN status.	31
566	5	Coefficient estimates across every specification and both samples.	32
567	6	Country-level staggered event-study, doubly-robust specifica-	
568		tion with log GDP per capita control. Shaded bands are point-	
569		wise 95% confidence intervals; relative periods beyond $\pm 12/9$	
570		are trimmed for having too few surviving cohorts to be infor-	
571		mative.	33

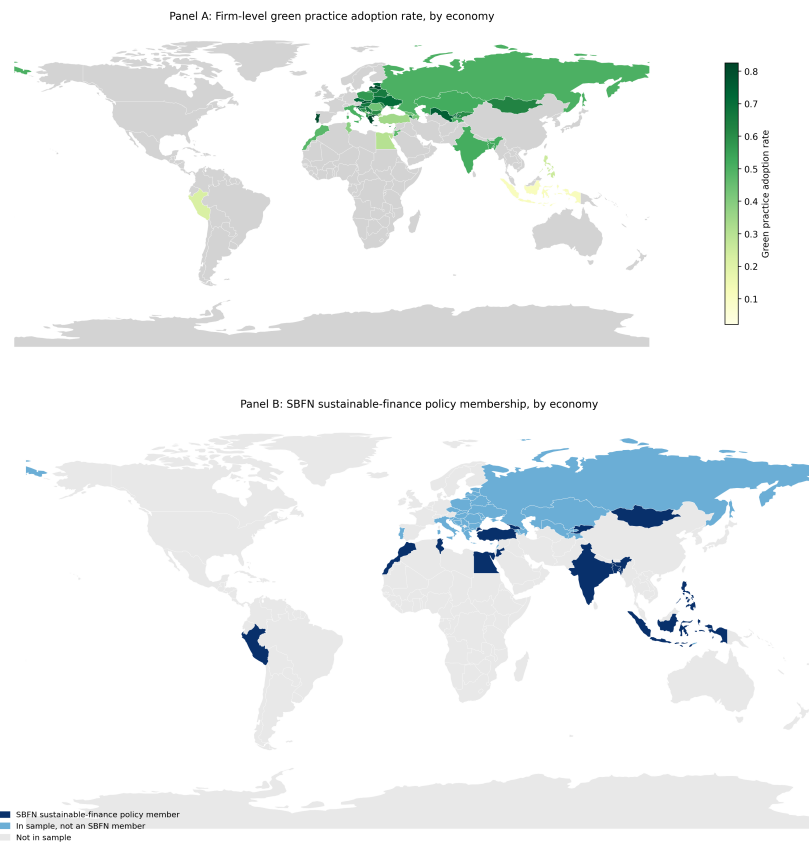


Figure 1: Green practice adoption and SBFN policy membership, by economy.

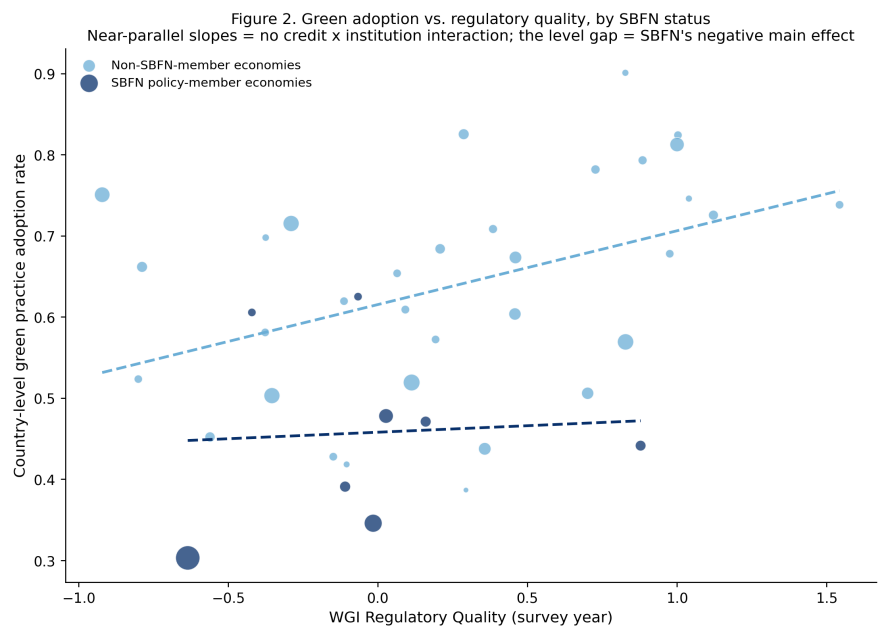


Figure 2: Green adoption vs. regulatory quality, by SBFN status.

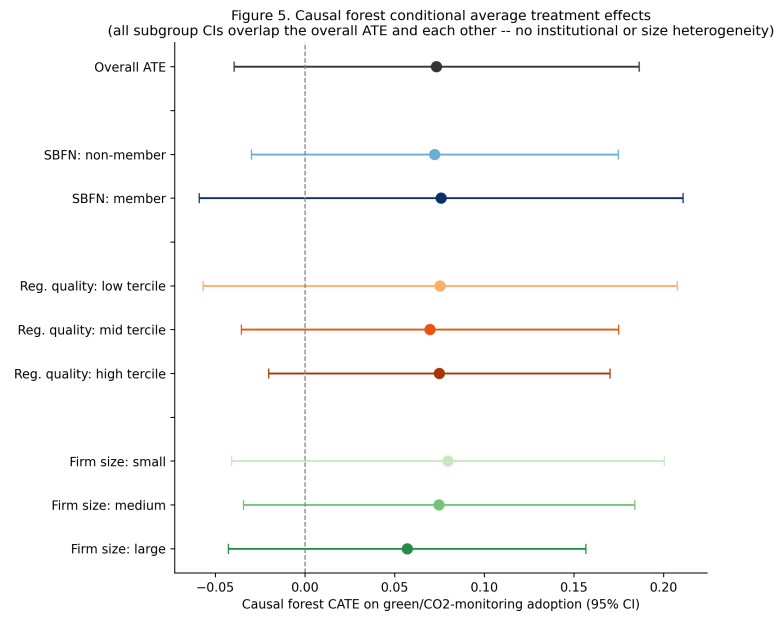


Figure 3: Causal forest conditional average treatment effects, by subgroup.

Figure 3. Combined sample coverage: 166 economies (41 primary rich-module + 162 global minimal-indicator, net of overlap)

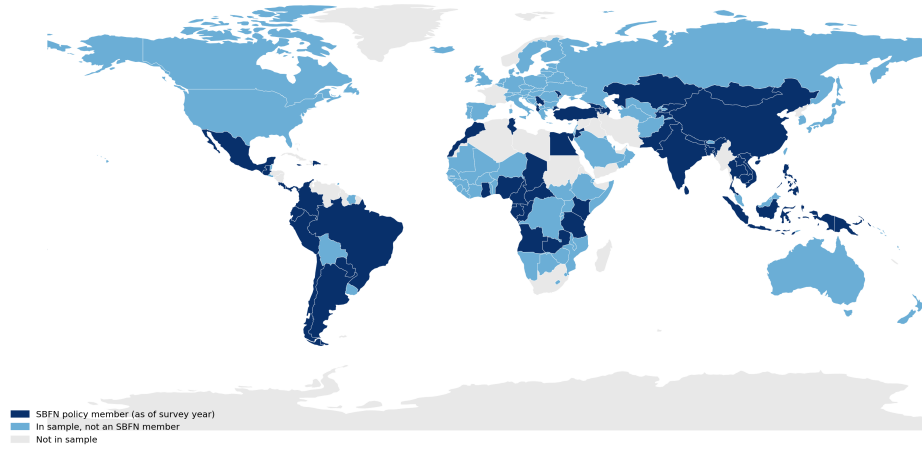


Figure 4: Combined sample coverage: 166 economies (41 primary + 162 global, net of overlap), shaded by SBFN status.

Figure 4: Coefficient estimates across every specification and both samples
(Panel A: robust and significant everywhere; Panels B-C: null everywhere)

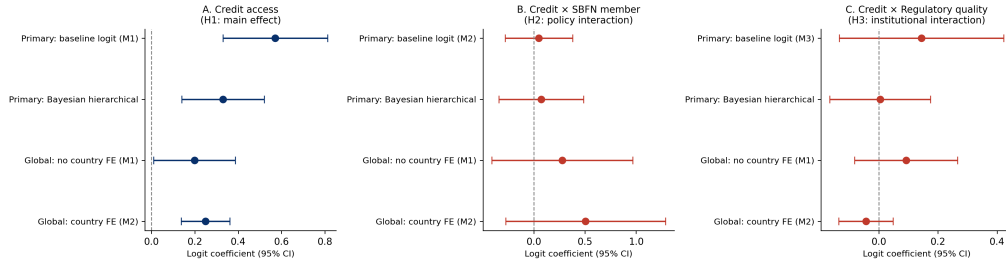


Figure 5: Coefficient estimates across every specification and both samples.

Country-level staggered event-study, SBFN adoption on macro environmental outcomes
 (Callaway-Sant'Anna, doubly-robust, log GDP per capita control; shaded = pointwise 95% CI;
 relative periods beyond $\pm 12/9$ trimmed, too few surviving cohorts to be informative)

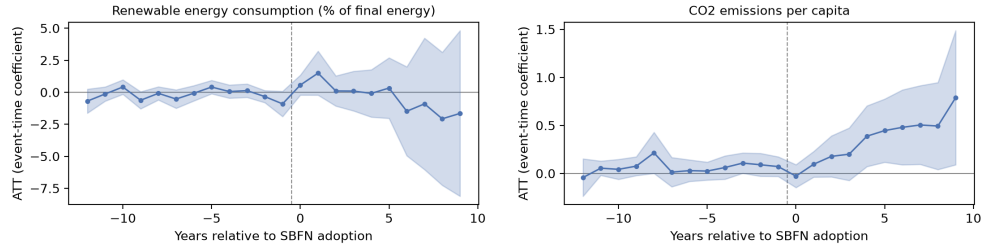


Figure 6: Country-level staggered event-study, doubly-robust specification with log GDP per capita control. Shaded bands are pointwise 95% confidence intervals; relative periods beyond $\pm 12/9$ are trimmed for having too few surviving cohorts to be informative.

Table 1: Baseline classical logit regressions (dependent variable: green practice adoption, binary). Country-clustered standard errors in parentheses.

	M1	M2	M3
Credit access	0.571*** (0.123)	0.459*** (0.090)	0.373*** (0.085)
SBFN member		-0.991*** (0.178)	-0.844*** (0.209)
Credit $\times$ SBFN		0.049 (0.168)	0.019 (0.140)
Regulatory quality			0.278 (0.174)
Credit $\times$ Reg. quality			0.144 (0.142)
SBFN $\times$ Reg. quality			0.053 (0.261)
Credit $\times$ SBFN $\times$ Reg. quality			-0.104 (0.272)
Log sales	0.061 (0.039)	0.041 (0.034)	0.068** (0.032)
Foreign owned	0.219 (0.143)	0.244* (0.139)	0.220 (0.143)
Exporter	0.537*** (0.092)	0.471*** (0.074)	0.412*** (0.079)
Sector, size controls	Yes	Yes	Yes
N	23,914	23,914	23,914
Pseudo R^2	0.058	0.089	0.093

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 2: Bayesian hierarchical model, posterior summary (four chains, 1,000 post-warmup draws each).

Parameter	Mean	SD	89% ETI low	89% ETI high	$\hat{R}$
Intercept	-1.93	0.246	-2.30	-1.50	1.00
Credit access	0.33	0.097	0.18	0.49	1.00
SBFN member	-0.57	0.300	-1.00	-0.09	1.01
Credit $\times$ SBFN	0.072	0.211	-0.26	0.40	1.00
Reg. quality (std.)	0.344	0.116	0.16	0.53	1.00
Credit $\times$ Reg. quality	0.004	0.087	-0.13	0.14	1.00

Table 3: Causal forest DML: average treatment effect, CATEs, and feature importances.

Estimate	Value	95% CI low	95% CI high
Overall ATE	0.0733	-0.0396	0.1862
CATE SBFN non-member	0.0723	-0.0300	0.1746
CATE SBFN member	0.0758	-0.0591	0.2107
CATE reg. quality tercile: low	0.0753	-0.0568	0.2075
CATE reg. quality tercile: mid	0.0697	-0.0355	0.1749
CATE reg. quality tercile: high	0.0749	-0.0203	0.1701
Feature	Importance		
SBFN member	0.0257		
Regulatory quality	0.1993		
Log sales	0.7562		
Exporter dummy	0.0189		

Table 4: Global-sample regressions (dependent variable: CO2-emissions monitoring, binary). Country-clustered standard errors in parentheses; M2's SBFN and regulatory-quality main effects are absorbed by country fixed effects and are not separately estimable.

	M1: no country FE	M2: country FE
Credit access	0.198** (0.096)	0.249*** (0.057)
SBFN member	-0.171 (0.184)	— (absorbed)
Credit $\times$ SBFN	0.278 (0.352)	0.506 (0.399)
Regulatory quality (std.)	0.155 (0.096)	— (absorbed)
Credit $\times$ Reg. quality	0.092 (0.089)	-0.044 (0.047)
Log sales	0.127*** (0.027)	0.288*** (0.028)
Foreign owned	0.621*** (0.083)	0.528*** (0.046)
Exporter	0.456*** (0.094)	0.335*** (0.044)
Country fixed effects	No	Yes
N	89,797	89,797
Countries	159	159

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Taiwan dropped (no WGI coverage).

Table 5: Country-level staggered event-study: overall average treatment effect of SBFN adoption on macro environmental outcomes. Callaway-Sant’Anna estimator, 999 bootstrap iterations; naive TWFE reported for comparison only. Full country-year panel, 2000–2024.

Outcome	Specification	ATT	SE	95% CI
Renewable energy %	Callaway-Sant’Anna, no covariates, never-treated	-1.192	0.802	[-2.763, 0.380]
Renewable energy %	Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated	0.674	0.916	[-1.122, 2.470]
Renewable energy %	Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated	0.653	0.919	[-1.147, 2.454]
Renewable energy %	Naive TWFE (post dummy, country+year FE)	-4.672***	1.127	[-6.882, -2.463]
CO2 per capita	Callaway-Sant’Anna, no covariates, never-treated	0.631***	0.179	[0.280, 0.982]
CO2 per capita	Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated	0.349**	0.156	[0.044, 0.655]
CO2 per capita	Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated	0.313**	0.148	[0.023, 0.603]
CO2 per capita	Naive TWFE (post dummy, country+year FE)	0.901***	0.204	[0.500, 1.302]

*** 95% CI excludes zero; ** 95% CI excludes zero, narrower margin. Underlying data and code in the replication package.